ZANKO ZANDSALIMI

University of Virginia

mfx2uq@virginia.edu **Charlottesville, Virginia, USA** **https://www.linkedin.com/in/zanko-zandsalimi/**
https://scholar.google.com/Zanko Zandsalimi

INTRODUCTION

I am a Ph.D. candidate in Civil and Environmental Engineering at the University of Virginia, specializing in machine learning for complex physical systems. My research focuses on developing scalable and reliable surrogate models using graph neural networks (GNNs), neural operators, and geometry-informed learning. I investigate how geometric heterogeneity, spatial connectivity, boundary conditions, and physical constraints can be incorporated into learning frameworks to improve generalization, computational efficiency, and stability in long-horizon predictions. My primary applications involve hydrologic and hydraulic simulations, particularly flood forecasting and inundation modeling, where nonlinear dynamics, interacting physical processes, and complex computational geometries pose significant challenges. My broader goal is to advance transferable, physics-consistent machine learning frameworks for efficient simulation and prediction across diverse physical systems.

EDUCATION

PhD in Civil Engineering (**GPA:** 4.00/4.00)
University of Virginia | Aug 2023 – Present | Virginia, USA
Project Title: Geometry-Informed Machine Learning for Scalable and Reliable Flood Simulation
Advisor: Negin Alemazkoor / **Co-advisor:** Majid Shafiee-Jood

---

M.Sc. in Civil Engineering (**GPA:** 4.00/4.00)
Tarbiat Modares University | 2016 – 2019 | Tehran, Iran
Thesis: Physics-Based Numerical simulation of Urban Flooding for Loss-of-Life Risk Assessment
Advisor: Seyed Ali Akbar Salehi Neyshabouri

RESEARCH INTERESTS

- Scientific Machine Learning
- Graph Neural Networks & Geometric Deep Learning
- Scalable & Stable Surrogate Modeling
- Physics-Informed & Geometry-Aware Learning
- Spatiotemporal Modeling & Forecasting
- Physics-Based Numerical & Hydrodynamic Modeling
- Flood Forecasting & Inundation Modeling
- Remote Sensing & GIS for Environmental Monitoring

PUBLICATIONS

Peer-Reviewed Publications

- Taghizadeh, M., **Zandsalimi, Z.**, Nabian, M. A., Goodall, J. L., Alemazkoor, N., "FloodForecaster: A Domain-Adaptive Geometry-Informed Neural Operator Framework for Rapid Flood Forecasting", *Journal of Hydrology*, 134512. DOI
- **Zandsalimi, Z.**, Barbosa, S. A., Alemazkoor, N., Goodall, J. L., & Shafiee-Jood, M. **(2025)**. "Deep Learning-Based Downscaling of Global Digital Elevation Models for Enhanced Urban Flood Modeling", *Journal of Hydrology*, 132687. DOI | GitHub
- Taghizadeh, M., **Zandsalimi, Z.**, Shafiee-Jood, M., & Alemazkoor, N. **(2025)**. Multi-fidelity graph neural networks for efficient and accurate flood hazard mapping. *Environmental Modelling & Software*, *106654*. DOI | GitHub
- Taghizadeh, M., **Zandsalimi, Z.**, Nabian MA., Shafiee-Jood, M., Alemazkoor, N., **(2025)**. "HydroGraphNet: Interpretable Physics-Informed Graph Neural Networks for Flood Forecasting", *Computer-Aided Civil and Infrastructure Engineering*, 1–21. DOI | GitHub
- Anand, H., Khayambashi, K., **Zandsalimi, Z.**, Taghizadeh, M., Hasnat, M. A., Alemazkoor, N., "Applications of Graph Neural Networks in Civil Infrastructures: A Review on Transportation, Power, Water, and Structural Systems", **preprint** in *Engineering Applications of Artificial Intelligence*, DOI.
- **Zandsalimi, Z.**, Feizabadi, S., Yazdi, J., & Salehi Neyshabouri, S. A. A. **(2024)**. Evaluating the Impact of Digital Elevation Models on Urban Flood Modeling: A Comprehensive Analysis of Flood Inundation, Hazard Mapping, and Damage Estimation. *Water Resources Management*, 1-26. DOI
- Sima, S., **Zandsalimi, Z.**, & Darzi, A. **(2023)**. A review and uncertainty analysis of Lake Urmia's hypsometric relationships. *Journal of Great Lakes Research*, 102239. DOI

Under Review:

- **Zandsalimi, Z.**, Taghizadeh, M., Shafiee-Jood, M., Alemazkoor, N., "Boundary-Aware Graph Neural Networks for Compound Flood Forecasting", **under review**.

- **Zandsalimi, Z.**, Taghizadeh, M., Lynn, S. L., Goodall, J. L., Shafiee-Jood, M., Alemazkoor, N., "Toward Stable Autoregressive Hydraulic Prediction in Stormwater Systems with Dual Node–Edge Updating Graph Neural Networks", **under review**
- Taghizadeh, M., **Zandsalimi, Z.**, Shafiee-Jood, M., Goodall, J. L., Alemazkoor, N., "Functional Generative Neural Operators for Fast Probabilistic Flood Inundation Forecasting", **under review**.

AWARDS AND HONORS

- **Virginia Sea Grant Graduate Research Fellowship**, 2026 – Awarded an $80,000 fellowship to investigate the economic value of flood monitoring and early warning systems in coastal Virginia, integrating AI-driven flood modeling and forecasting to support emergency management and decision-making. Link
- **Olsen Graduate Fellowship**, University of Virginia, 2025 – Awarded to top engineering Ph.D. students for academic and research excellence; additional $12,000 stipend.
- **Virginia Lakes and Watersheds Association (VLWA) Scholarship**, 2025.
- **Environmental Futures Fellowship**, University of Virginia, 2025. Link
- **First Place**, CEE Spring Graduate Symposium, UVA, 2025.
- **Link Lab Student Flash Talks Award**, University of Virginia, 2025.
- **Provost Fellowship**, University of Virginia – Graduate student research support.
- **Top Ranked Student**, M.Sc., Tarbiat Modares University.
- **Top Ranked Student**, B.Sc., University of Kurdistan.
- **1st Prize – WaPOR Hackathon: From Database to Application (2020)**. International competition with 241 participants from 57 countries, organized by IHE Delft, Wageningen University, and MetaMeta.

PRESENTATIONS

- **Zandsalimi, Z.**, Taghizadeh, M., Shafiee-Jood, M., Alemazkoor, N., *Learning Boundary–Interior Coupling for Compound Flood Forecasting Using Graph Neural Networks*. Poster Presentation, Virginia Crossroads: Flood Resilience across the Commonwealth Symposium 2026, Charlottesville, VA, **Mar 2026**.
- **Zandsalimi, Z.**, Taghizadeh, M., Shafiee-Jood, M., Alemazkoor, N., *Learning Boundary–Interior Coupling for Compound Flood Forecasting Using Graph Neural Networks*. Poster Presentation, CEE Graduate Symposium, University of Virginia, **Feb 2026**.
- **Zandsalimi, Z.**, Shafiee-Jood, M., Alemazkoor, N., *Spatiotemporal Graph Neural Network-Based Surrogate for Compound Flood Forecasting*. Poster Presentation, AGU Fall Meeting 2025, New Orleans, LA, **Dec 2025**.
- Taghizadeh, M., **Zandsalimi, Z.**, Nabian, M. A., Goodall, J. L., Alemazkoor, N., *Domain-Adaptive Geometry-Informed Neural Operators for Rapid Flood Forecasting*. Poster Presentation, AGU Fall Meeting, New Orleans, LA, **Dec 2025**.
- **Zandsalimi, Z.**, Climate Risk and Resilience – Urban Hydrology Group, FM Global. Presentation on research conducted during summer internship. Norwood, MA, **Aug 2025**.
- **Zandsalimi, Z.**, Shafiee-Jood, M., Alemazkoor, N., *End-to-End Graph Neural Networks for Real-Time Hydraulic Prediction in Stormwater Systems*. Oral Presentation, Engineering Mechanics Institute Conference (EMI 2025), University of California, Irvine (Anaheim, CA), **May 27–30, 2025**.
- Taghizadeh, M., **Zandsalimi, Z.**, Shafiee-Jood, M., Alemazkoor, N., *Interpretable Physics-Informed Graph Neural Networks for Flood Forecasting*. Oral Presentation, Engineering Mechanics Institute Conference (EMI 2025), University of California, Irvine (Anaheim, CA), **May 27–30, 2025**.
- **Zandsalimi, Z.**, Barbosa, S. A., Goodall, J. L., Alemazkoor, N., Shafiee-Jood, M., *From Global to Local: Enhancing Urban DEMs Using Deep Learning and Integration of Remote Sensing Imagery and Global Elevation Data*. Poster Presentation, AGU Fall Meeting, Washington, D.C., **Dec 2024**.
- **Zandsalimi, Z.**, Barbosa, S. A., Alemazkoor, N., Goodall, J. L., Shafiee-Jood, M., *Deep Learning-Based Downscaling of DEMs for Urban Flood Modeling*. UVA Engineering Poster and Flash Talk, Charlottesville, VA, **2024**.

RESEARCH EXPERIENCES

Graduate Research Assistant
University of Virginia | Aug 2023 – Present | Virginia, USA

- Develop graph neural network and neural operator surrogates for large-scale spatiotemporal physical simulation, focusing on transferability across geometries, stable long-horizon autoregressive prediction, and computational efficiency, with models trained on GPU-accelerated HPC clusters.
- Co-developed, in collaboration with NVIDIA, **HydroGraphNet**, an interpretable physics-informed GNN released in NVIDIA PhysicsNeMo, and **FloodForecaster**, a domain-adaptive geometry-informed neural operator for rapid forecasting on new domains.
- Developed boundary-aware GNNs for compound flood forecasting and dual node–edge updating GNNs for stable autoregressive prediction in interconnected stormwater networks; contributed to multi-fidelity GNNs and generative neural operators for probabilistic forecasting.
- Developed deep learning (U-Net) methods for DEM downscaling with transfer learning and domain adaptation, improving terrain inputs for physical simulation in data-scarce regions.

Summer Research Intern
FM Global – Climate Risk and Resilience Group | Jun 2025 – Aug 2025 | Norwood, MA, USA

- Developed a scalable, automated Python pipeline for terrain correction in hydraulic modeling, detecting road–river intersections, extracting culvert geometry, and applying targeted DEM corrections.
- Built an interactive dashboard for analyzing terrain refinements and derived stream networks; documented the workflow in an internal technical report and presented results to the research team.

Research Assistant and Water Resources Engineer
Tarbiat Modares University | 2016 – 2022 | Tehran, Iran

- Built physics-based 2D hydrodynamic models (HEC-RAS, MIKE FLOOD) for urban flood simulation, hazard mapping, and loss-of-life risk assessment in Tehran (M.Sc. research).
- Contributed to applied projects in flood risk assessment, SWAT-based rainfall–runoff modeling with remote-sensing inputs, and a beta Web-GIS platform for ecosystem management.

TECHNICAL SKILLS

- **Programming & Scientific Computing:** Python, MATLAB, NumPy, SciPy, Pandas
- **Machine Learning & Deep Learning:** PyTorch, PyTorch Geometric, Scikit-learn, Graph Neural Networks, Neural Operators, CNNs
- **High-Performance Computing:** GPU-Accelerated Training, HPC Clusters, Scientific Workflow Automation
- **Geospatial Analysis & Remote Sensing:** ArcGIS, QGIS, GeoPandas, Rasterio, Google Earth Engine
- **Hydrologic & Hydraulic Modeling:** HEC-RAS, SWMM, MIKE FLOOD, SWAT
- **Data Visualization & Dashboards:** Matplotlib, Plotly, Dash

TEACHING & MENTORSHIP EXPERIENCE

Instructor / Mentor
University of Virginia | 2025 – Fall Semester | Virginia, USA

- Instructor for the NSF-funded program "Empowering High School Teachers with AI" (PI: Dr. Somayeh Asadi). Delivered hands-on training in Python programming, data analysis, and introductory artificial intelligence for high school STEM teachers.

Teaching Assistant
Tarbiat Modares University | 2018 – 2022 (4 semesters) | Tehran, Iran

- Graduate teaching assistant for the course "RS and GIS Applications in Environmental Monitoring." Responsibilities included leading tutorials, assisting graduate students with lab exercises, grading assignments, and providing academic support. (Advisor: Dr. Somayeh Sima)